Andrew Koh

Graduate Mechatronics Engineer
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Experience

Production and Manufacturing Assistant Aug. 2022 – Current
Agile Workspace Ltd. *Auckland, New Zealand*

- **LLM & Workflow Automation:** Built an end-to-end automated rostering pipeline using a Facebook Messenger chatbot and LLM API, eliminating manual manager entry and reducing scheduling overhead per update to near zero.
- **Embedded Systems Development:** Developed an occupancy sensor application on an ATMega MCU using accelerometer and PIR sensors, achieving an 87% reduction in detection time.
- **Hardware Manufacturing & QA:** Manufacturing, documentation, and integration-testing of diverse electronic hardware including embedded devices, network controllers, and mini PCs over various communication protocols including: Wi-Fi, radio, and Ethernet.

Firmware Engineer Intern Nov 2023 - Feb 2024
Resideo *Auckland, New Zealand*

- **Embedded Driver Development:** Designed and implemented peripheral drivers on an ARM Cortex-M0+ microcontroller to control a motorized camera lens, allowing precise hardware-level actuation.
- **SoC-MCU Communication:** Defined and extended a custom I2C protocol with 6 new lens control commands and enhanced the SoC CLI, enabling end-to-end MCU control through the host interface.
- **Autofocus Algorithm:** Designed a contrast-based autofocus algorithm using a finite state machine, significantly reducing lens traversal distance compared to a full-range sweep.

Projects

Final Year Project — 6D Pose Estimation for Robotic Assembly Mar 2024 – Oct 2024

- **3D Computer Vision:** Implemented 6D pose estimation algorithms using a depth camera for real-time point cloud acquisition, achieving less than 1mm of accuracy error in on target objects.
- **Motion Control:** Applied forward and inverse kinematics to translate pose estimates into joint-space commands, enabling the robotic arm to position its end-effector to target objects.
- **ROS Integration:** Developed the full pipeline within the ROS framework for robust sensor fusion and processing.

Low-latency BLE MIDI Musical Instrument Controller May 2018 – May 2020

- **RTOS Architecture:** Architected a dual-core firmware solution isolating high-frequency hardware scanning on Core 1 from the BLE stack on Core 0, ensuring CPU availability for time-critical inputs .
- **High Performance Polling** Achieved sub-millisecond input latency via deterministic peripheral scanning using timer hardware interrupts.
- **Automated Testing:** Improved reliability and testability by implementing an automated unit testing framework.

Tele-Oximeter | *React, Flask, Supabase* June 2025 - Nov 2025

- **IoT Pipeline Design:** Designed an edge-to-cloud pipeline streaming BLE pulse oximeter data through a REST API to a React dashboard for real-time remote SpO_2 and pulse monitoring.
- **Multi-user Session Management:** Built a session key system supporting concurrent users, isolating each device's data stream with a simple dashboard access mechanism.
- **End-to-End Testing:** Implemented UI testing with Playwright, validating session key entry, real-time data rendering, and live chart updates from a simulated data stream.

Education

University of Auckland Auckland, New Zealand
Bachelor of Engineering (Honours) - Mechatronics, First Class Honours | GPA: 7.7/9 *2021 - 2024*

Technical Skills

Languages: Python, C/C++
Hardware/Architecture: AVR, ARM, RISC-V
Frameworks: ROS, Flask, React, FreeRTOS, Unity Testing, Playwright, Platform IO
Robotics: Control Theory, Rapid Prototyping, Omron PLC
CAD: Inventor, Onshape